# Supervised Learning: Exercise of Information Theory Part 1

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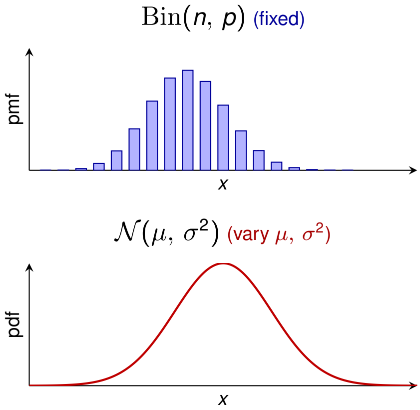
May 21, 2026

# Exercise 1: Kullback-Leibler Divergence

(a) Approximate the binomial $\mathrm{Bin}(n, p)$ with a Gaussian $\mathcal{N}(\mu, \sigma^2)$. To find a suitable fit, investigate the Kullback-Leibler divergence (KLD) in terms of $\boldsymbol{\theta} = (\mu, \sigma^2)^T$.

1. Write down the KLD for the given setup.
2. Derive the gradients w.r.t. $\boldsymbol{\theta}$.
3. Is there an analytic solution for the optimal parameter setting? If yes, derive the corresponding solution. If no, give a short reasoning.
4. Independent of the previous exercise, state a numerical procedure to minimize the KLD.




Fix $\mathrm{Bin}(n, p)$; tune $\mu, \sigma^2$ of $\mathcal{N}$ to fit it.

# Solution to Question (a.1)

Question: Write down the KLD for the given setup.

- Let $f$ be pmf of $\mathrm{Bin}(n, p)$; we have $\mathbb{E}_f[X] = np$ and $\mathrm{Var}_f[X] = np(1 - p)$.
- Let $q$ be the density of $\mathcal{N}(\mu, \sigma^2)$.

**Part (a) roadmap**

1. **Write the KLD between binomial $f$ and Gaussian $q$**
2. Compute gradients of the KLD w.r.t. $\boldsymbol{\theta}$
3. Find critical points and check the Hessian
4. State a numerical procedure to minimize the KLD

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- Let $q$ be the density of $\mathcal{N}(\mu, \sigma^2)$.

The KL divergence between $f$ and $q$ is

$$
\begin{aligned}
D_{KL}(f \,||\, q) &= \mathbb{E}_f\left[\log \frac{f(X)}{q(X;\boldsymbol{\theta})}\right] \\
&= \underbrace{\mathbb{E}_f\left[\log f(X)\right]}_{c_1} - \mathbb{E}_f[\log q(X|\boldsymbol{\theta})] \\
&= c_1 - \mathbb{E}_f\left[\log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{1}{2\sigma^2}(X-\mu)^2\right)\right)\right] \\
&= c_1 - \mathbb{E}_f\left[-\underbrace{\tfrac{1}{2}\log(2\pi)}_{c_2} - \tfrac{1}{2}\log(\sigma^2) - \tfrac{1}{2\sigma^2}(X-\mu)^2\right]
\end{aligned}
$$

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# Solution to Question (a.1): Continued

$$
\begin{aligned}
D_{KL}(f \,||\, q) &= \underbrace{c_1 + c_2}_{c_3} + \tfrac{1}{2}\log(\sigma^2) + \tfrac{1}{2\sigma^2}\mathbb{E}_f[(X-\mu)^2] \\
&= c_3 + \log(\sigma) + \tfrac{1}{2\sigma^2}\left(\underbrace{\mathbb{E}_f[X^2]}_{\mathrm{Var}_f[X]+\mathbb{E}_f[X]^2} - 2\mu\mathbb{E}_f[X] + \mu^2\right) \\
&= c_3 + \log(\sigma) + \tfrac{1}{2\sigma^2}(np(1-p) + (np)^2 - 2\mu np + \mu^2) \\
&= c_3 + \log(\sigma) + \tfrac{1}{2\sigma^2}(np(1-p) + (np-\mu)^2)
\end{aligned}
$$

**Part (a) roadmap**

1. **Write the KLD between binomial $f$ and Gaussian $q$**
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# Solution to Question (a.2)

Question: Derive the gradients w.r.t. $\boldsymbol{\theta}$.

$$\frac{\partial D_{KL}(f \,||\, q)}{\partial \mu} = \frac{\partial}{\partial \mu} \frac{(np-\mu)^2}{2\sigma^2} = \frac{2(np-\mu)\cdot(-1)}{2\sigma^2} = \frac{\mu - np}{\sigma^2}$$

$$\begin{aligned}\frac{\partial D_{KL}(f \,||\, q)}{\partial \sigma^2} &= \frac{\partial}{\partial \sigma^2}\left(\tfrac{1}{2}\log(\sigma^2) + \tfrac{1}{2\sigma^2}(np(1-p) + (np-\mu)^2)\right) \\ &= \tfrac{1}{2\sigma^2} - \tfrac{1}{2(\sigma^2)^2}(np(1-p) + (np-\mu)^2) \\ &= \tfrac{1}{2\sigma^2} - \tfrac{1}{2\sigma^4}\mathbb{E}_f[(X-\mu)^2]\end{aligned}$$

**Part (a) roadmap**

1. Write the KLD between binomial $f$ and Gaussian $q$

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**Part (a) roadmap**

1. Write the KLD between binomial $f$ and Gaussian $q$
2. **Compute gradients of the KLD w.r.t. $\boldsymbol{\theta}$**
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# Solution to Question (a.3)

Question: Is there an analytic solution for the optimal parameter setting? If yes, derive the corresponding solution. If no, give a short reasoning.

Solution: Yes, there is. By setting the partial derivatives to zero we can identify the critical points.

$$\frac{\partial D_{KL}(f \,||\, q)}{\partial \mu} = 0 \Leftrightarrow \hat{\mu} = np = \mathbb{E}_f[X]$$

$$\frac{\partial D_{KL}(f \,||\, q)}{\partial \sigma^2} = 0 \Leftrightarrow \frac{1}{2\hat{\sigma}^2} = \frac{1}{2\hat{\sigma}^4}\mathbb{E}_f[(X - \hat{\mu})^2]$$

$$\Leftrightarrow \hat{\sigma}^2 = \mathbb{E}_f[(X - \mathbb{E}_f[X])^2] = \mathrm{Var}_f[X] = np(1-p)$$

To prove this critical point is a minimum, check that the Hessian w.r.t. $(\mu, \sigma^2)$ is positive definite at $(\hat{\mu}, \hat{\sigma}^2)$.

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# Solution to Question (a.3): Continued

The Hessian matrix is

$$\nabla^2 = \begin{pmatrix} \frac{1}{\sigma^2} & (np-\mu)\frac{1}{\sigma^4} \\ (np-\mu)\frac{1}{\sigma^4} & -\frac{1}{2\sigma^4} + \frac{1}{\sigma^6}\mathbb{E}_f[(X-\mu)^2] \end{pmatrix}$$

Evaluated at $(\hat{\mu}, \hat{\sigma}^2)$, we obtain

$$\nabla^2\Big|_{(\hat{\mu},\hat{\sigma}^2)} = \begin{pmatrix} \frac{1}{np(1-p)} & 0 \\ 0 & \frac{1}{2(np(1-p))^2} \end{pmatrix} = \begin{pmatrix} \frac{1}{\mathrm{Var}_f(X)} & 0 \\ 0 & \frac{1}{2\mathrm{Var}_f(X)^2} \end{pmatrix}$$

So the Hessian at $(\hat{\mu}, \hat{\sigma}^2)$ is P.S.D.

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# Solution to Question (a.4)

Question: State a numerical procedure to minimize the KLD.

**Gradient descent** on the KLD: iterate

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \nabla_{\boldsymbol{\theta}} D_{KL}(f \,\|\, q_{\boldsymbol{\theta}_t})$$

from $\boldsymbol{\theta}_0$ with step size $\eta > 0$.

Components (from (a.2)):

- $\partial D_{KL}/\partial \mu = (\mu - np)/\sigma^2$
- $\partial D_{KL}/\partial \sigma^2 = \frac{1}{2\sigma^2} - \frac{1}{2\sigma^4} \mathbb{E}_f[(X-\mu)^2]$

*Remark:* (a.3) yields a closed form; gradient descent is the fallback when no analytic solution exists.

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1. Write the KLD between binomial $f$ and Gaussian $q$
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# Question (b), (c) and (d)

(b) Sample points according to the true distribution and visualize the KLD for different parameter settings of the Gaussian distribution (including the optimal one if available).

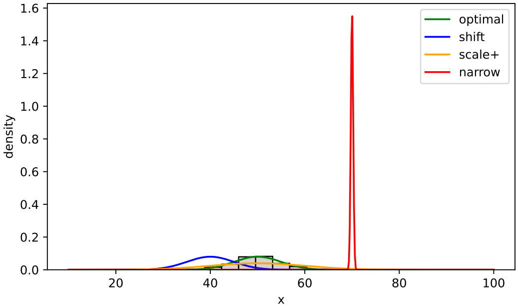
(c) Create a surface plot with axes $n$ and $p$ and colour value equal to the KLD for the optimal normal distribution.

(d) Based on the previous result,

1. how can the behaviour for varying $p$ be explained?
2. how can the behaviour for varying $n$ be explained?

# Solution to Question (b)

```python
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

# Setup: Bin(n=100, p=0.5), draw B=1000 samples
n, p, B = 100, 0.5, 1000
X = stats.binom.rvs(n, p, size=B, random_state=0)

# Optimal Gaussian moments (match the binomial)
mu0, sd0 = n*p, np.sqrt(n*p*(1-p))
xs = np.linspace(10, 100, 500)

# 4 candidate Gaussians: (mean, sd, color, label)
pdfs = [(mu0,       sd0,      'green',  'optimal'),
        (mu0 - 10, sd0,      'blue',   'shift'),
        (mu0,       2*sd0,    'orange', 'scale+'),
        (mu0 + 20, p*(1-p), 'red',    'narrow')]

# Histogram of samples, overlay each candidate density
fig, ax = plt.subplots(figsize=(7, 4))
ax.hist(X, bins=25, range=(10, 100), density=True,
        color='lightgray', edgecolor='black')
for m, s, c, lbl in pdfs:
    ax.plot(xs, stats.norm.pdf(xs, m, s), color=c, label=lbl)
ax.set_xlabel('x'); ax.set_ylabel('density'); ax.legend()
plt.savefig('ex_13_b.pdf', bbox_inches='tight')

# KLD (up to additive constant) from the closed form
def kld(mu, s2):
    return 0.5*np.log(s2) + 0.5/s2 * (n*p*(1-p) + (n*p - mu)**2)

for m, s, _, lbl in pdfs:
    print(f'{lbl:8s}: {kld(m, s**2):.4f}')
```

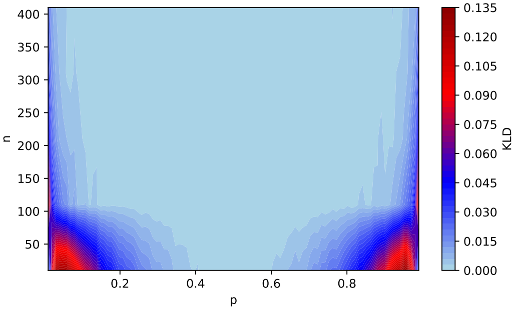



KLD up to additive constant:

```
optimal:  2.11  shift:  4.11
scale+:  2.43  narrow:  3398.61
```

# Solution to Question (c)

```python
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
from matplotlib.colors import LinearSegmentedColormap

# Grids over (p, n); B = MC sample size per cell
p_seq = np.linspace(0.01, 0.99, 100)
n_seq = np.arange(10, 510, 100)
rng = np.random.default_rng(0)

# Approx KLD by Monte Carlo: E_f[log f - log q] over B draws
def kld_approx(n, p, B=10000):
    x = rng.binomial(n, p, size=B)
    lf = stats.binom.logpmf(x, n, p)
    lq = stats.norm.logpdf(x, n*p, np.sqrt(n*p*(1-p)))
    return max(np.mean(lf - lq), 0)  # clamp tiny negatives

# Evaluate KLD at each (n, p) grid point
Z = np.array([[kld_approx(n_, p_) for n_ in n_seq]
              for p_ in p_seq])

# Custom blue->red colormap (matches the R original)
cmap = LinearSegmentedColormap.from_list(
    'c', ['lightblue', 'blue', 'red', 'darkred'])

# Filled-contour heatmap of the KLD over (p, n)
fig, ax = plt.subplots(figsize=(7, 4))
cs = ax.contourf(p_seq, n_seq, Z.T, levels=50, cmap=cmap)
fig.colorbar(cs, ax=ax, label='KLD')
ax.set_xlabel('p'); ax.set_ylabel('n')
plt.savefig('ex_13_c.pdf', bbox_inches='tight')
```

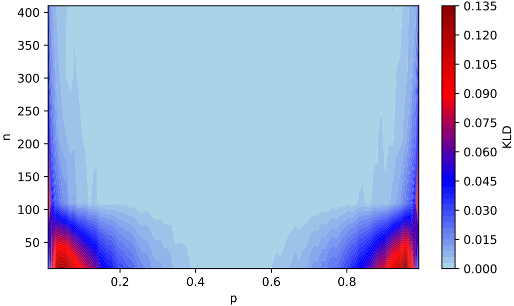


# Solution to Question (d)

**KLD over the $(n, p)$ grid:**

- Near zero for most $(n, p)$.
- Large near the edges $p \to 0$ or $p \to 1$ (skewed binomial).
- Worsens for small $n$ (CLT hasn't kicked in).

$\Rightarrow$ Gaussian fits the binomial poorly exactly when the binomial is highly skewed or under-sampled.



# Exercise 2: The Convexity of KL Divergence

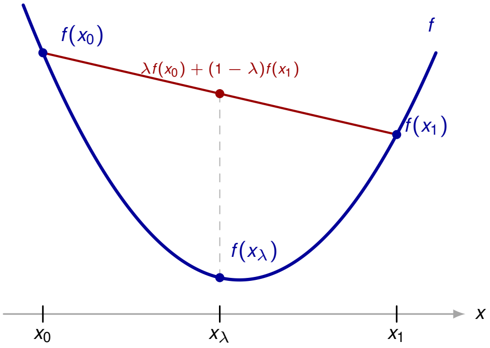
Let $p$ and $q$ be the PDFs of a pair of absolutely continuous distributions.
(a) Prove that the KL divergence is convex in the pair $(p, q)$, i.e.,

$$D_{KL}\left(\lambda p_1 + (1-\lambda)p_2 \,||\, \lambda q_1 + (1-\lambda)q_2\right) \leq \lambda D_{KL}\left(p1 \,||\, q_1\right) + (1-\lambda)D_{KL}\left(p_2 \,||\, q_2\right),$$

where $(p_1, q_1)$ and $(p_2, q_2)$ are two pairs of distribution and $0 \leq \lambda \leq 1$.

Hint: you can use the log sum inequality, namely that
$(a_1 + a_2)\log\left(\frac{a_1+a_2}{b_1+b_2}\right) \leq a_1 \log\frac{a_1}{b_1} + a_2 \log\frac{a_2}{b_2}$ holds for $a_1, a_2, b_1, b_2 \geq 0$.

# Convexity: A 1D Picture




For $\lambda \in [0, 1]$, let $x_\lambda = \lambda x_0 + (1 - \lambda)x_1$.
Convexity of $f$ means:

$$f(x_\lambda) \le \lambda f(x_0) + (1 - \lambda)f(x_1).$$

- $x_\lambda$ is a convex combination of $x_0$ and $x_1$.
- $\lambda f(x_0) + (1 - \lambda)f(x_1)$ is the chord value at $x_\lambda$.
- The chord lies above the curve.

In Exercise 2:

- Each “$x$” is a pair $(p, q)$ of distributions.
- $f$ is the KL divergence $D_{KL}(p \,\|\, q)$.

# Solution to Question (a)

$$D_{KL}\left(\lambda p_1 + (1-\lambda)p_2 \,||\, \lambda q_1 + (1-\lambda)q_2\right)$$

# Solution to Question (a)

$$
\begin{aligned}
& D_{KL}\left(\lambda p_1 + (1-\lambda) p_2 \;||\; \lambda q_1 + (1-\lambda) q_2\right) \\
&= \int_{\mathcal{X}} \left( (\lambda p_1(x) + (1-\lambda) p_2(x)) \log \frac{\lambda p_1(x) + (1-\lambda) p_2(x)}{\lambda q_1(x) + (1-\lambda) q_2(x)} \right) \mathrm{d}x
\end{aligned}
$$

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&\leq \int_{\mathcal{X}} \left( \lambda p_1(x) \log \frac{p_1(x)}{q_1(x)} + (1-\lambda)p_2(x) \log \frac{(1-\lambda)p_2(x)}{(1-\lambda)q_2(x)} \right) \mathrm{d}x \quad \text{(Use hint)}
\end{aligned}
$$

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&\leq \int_{\mathcal{X}} \left( \lambda p_1(x) \log \frac{p_1(x)}{q_1(x)} + (1-\lambda)p_2(x) \log \frac{(1-\lambda)p_2(x)}{(1-\lambda)q_2(x)} \right) \mathrm{d}x \quad \text{(Use hint)} \\
&= \lambda \int_{\mathcal{X}} \left( p_1(x) \log \frac{p_1(x)}{q_1(x)} \right) \mathrm{d}x + (1-\lambda) \int_{\mathcal{X}} \left( p_2(x) \log \frac{p_2(x)}{q_2(x)} \right) \mathrm{d}x
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&\le \int_{\mathcal{X}} \left( \lambda p_1(x) \log \frac{p_1(x)}{q_1(x)} + (1-\lambda)p_2(x) \log \frac{(1-\lambda)p_2(x)}{(1-\lambda)q_2(x)} \right) \mathrm{d}x \quad \text{(Use hint)} \\
&= \lambda \int_{\mathcal{X}} \left( p_1(x) \log \frac{p_1(x)}{q_1(x)} \right) \mathrm{d}x + (1-\lambda) \int_{\mathcal{X}} \left( p_2(x) \log \frac{p_2(x)}{q_2(x)} \right) \mathrm{d}x \\
&= \lambda D_{KL}\left(p_1 \,||\, q_1\right) + (1-\lambda) D_{KL}\left(p_2 \,||\, q_2\right).
\end{aligned}
$$